
CSCI 0410 / 144

— Professor Amy Greenwald —

FOUNDATIONS OF AI & ML

Fall 2026

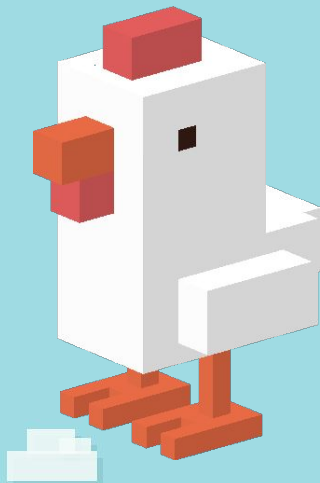
CS610410

A broad coverage of core topics in AI as a prelude to students taking more in-depth AI courses later on.

Students will study key AI models and machine learning to build socially responsible AI.

Professor

Amy Greenwald



I've been a Brown prof for longer than most of the students in this class have been alive. Throughout my career, I've taught intro AI and intro CS (with the aim of teaching students to program functionally, not dysfunctionally), Internet Agent Economics, and Algorithmic Game Theory. My research area is AI, specifically multi-agent systems, which concerns autonomous agent and system design for effective agent interaction. On the rare occasion when I'm not busy at work, I enjoy sports: soccer, biking, tennis, ice hockey, and more!

HTAs

Salman Aji



Mehdi Atmani



STA
A

Myan Nguyen



UTAs

Ashley Luo



David Vassalluzzo



Joanne Ding



UTAs

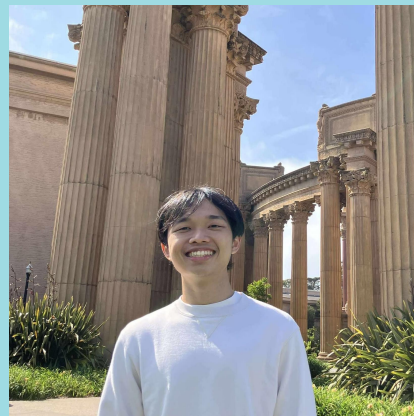
John Lockwood



Sydney Rondeau



Tyrone Serapio

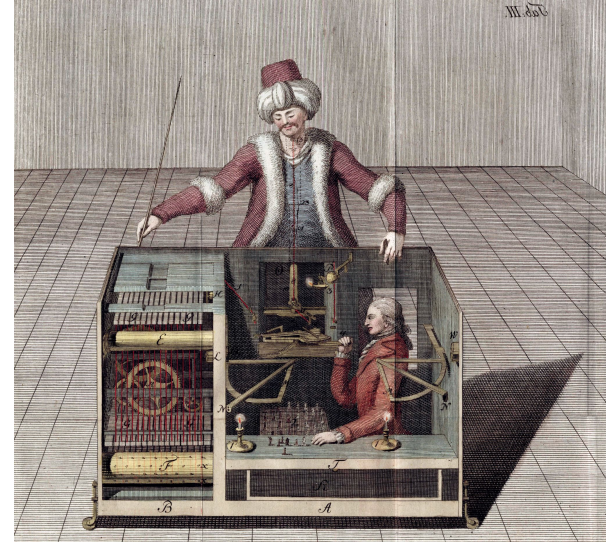




AI History



AI in Antiquity



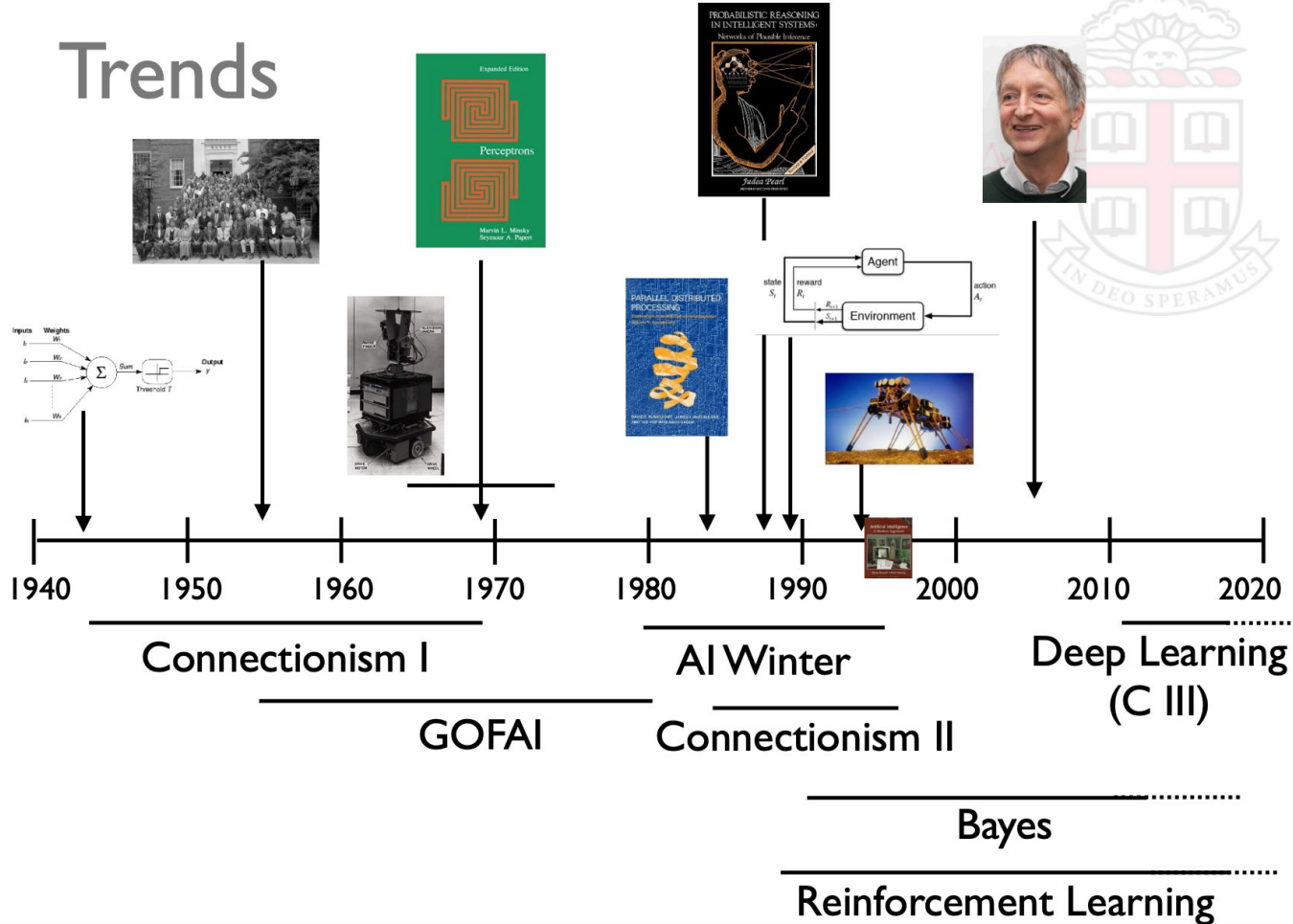
Sources: Wikipedia



AI History

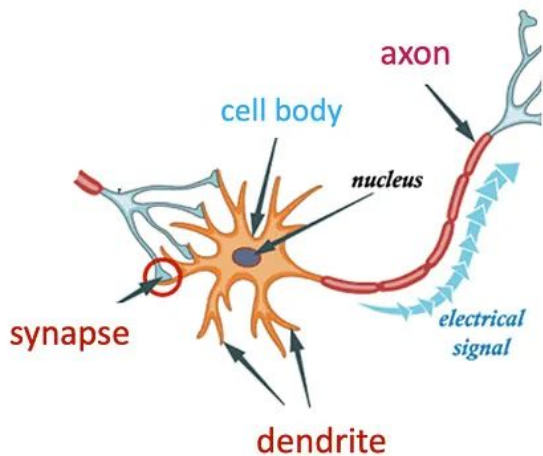


Trends

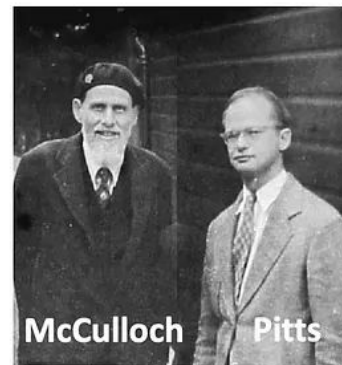
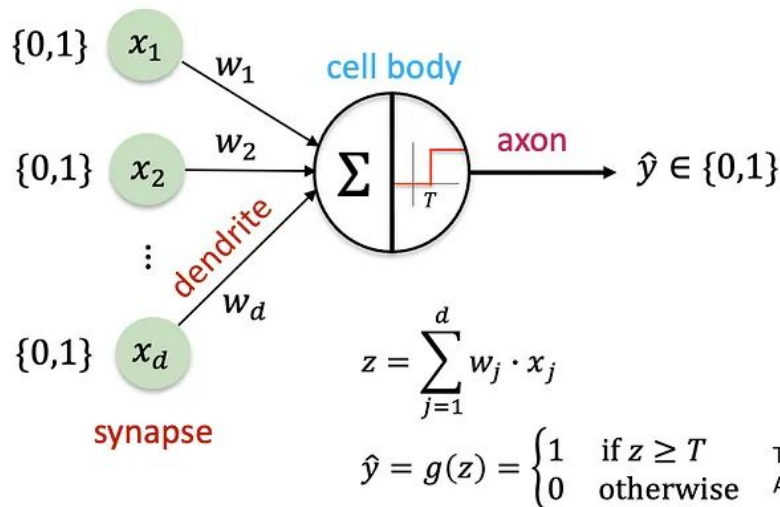


1943: McCulloch & Pitts

A Biologic Neuron



A McCulloch-Pitts Neuron



Source: [Medium](https://medium.com/@mcculloch-pitts)

1948-50: Claude Shannon, (Arguably) The Father of AI

The Bell System Technical Journal

Vol. XXVII

July, 1948

No. 3

A Mathematical Theory of Communication

By C. E. SHANNON

INTRODUCTION

THE recent development of various methods of modulation such as PCM and PPM which exchange bandwidth for signal-to-noise ratio has intensified the interest in a general theory of communication. A basis for such a theory is contained in the important papers of Nyquist¹ and Hartley² on this subject. In the present paper we will extend the theory to include a number of new factors, in particular the effect of noise in the channel, and the savings possible due to the statistical structure of the original message and due to the nature of the final destination of the information.

The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. Frequently the messages have *meaning*; that is they refer to or are correlated according to some system with certain physical or conceptual entities. These semantic aspects of communication are irrelevant to the engineering problem. The significant aspect is that the actual message is one selected from a set of possible messages. The system must be designed to operate for each possible selection, not just the one which will actually be chosen since this is unknown at the time of design.

Philosophical Magazine, Ser.7, Vol. 41, No. 314 - March 1950.

XXII. Programming a Computer for Playing Chess¹

By CLAUDE E. SHANNON

Bell Telephone Laboratories, Inc., Murray Hill, N.J.²

[Received November 8, 1949]

1. INTRODUCTION

This paper is concerned with the problem of constructing a computing routine or "program" for a modern general purpose computer which will enable it to play chess. Although perhaps of no practical importance, the question is of theoretical interest, and it is hoped that a satisfactory solution of this problem will act as a wedge in attacking other problems of a similar nature and of greater significance. Some possibilities in this direction are: -

- (1) Machines for designing filters, equalizers, etc.
- (2) Machines for designing relay and switching circuits.
- (3) Machines which will handle routing of telephone calls based on the individual circumstances rather than by fixed patterns.
- (4) Machines for performing symbolic (non-numerical) mathematical operations.
- (5) Machines capable of translating from one language to another.
- (6) Machines for making strategic decisions in simplified military operations.
- (7) Machines capable of orchestrating a melody.
- (8) Machines capable of logical deduction.

1950: Alan Turing, The Father of Computer Science



VOL. LIX. No. 236.]

[October, 1950

MIND

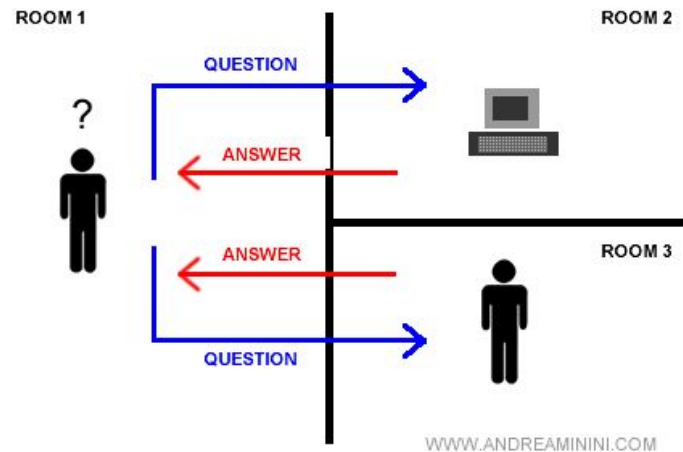
A QUARTERLY REVIEW
OF
PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning



The Turing Test

1956: Dartmouth Conference



1956: Dartmouth Conference

“The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. ...

An attempt will be made to find out how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. ...

We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.”

1956: Dartmouth Conference

The workshop fell short of its expectations.

But Newell & Simon's Logical Theorist managed to prove all theorems in Chapter 2 of Russell & Whitehead's *Principia Mathematica*.

Newell & Simon submitted a paper, co-authored with the Logical Theorist, to the Journal of Symbolic Logic. It was rejected.

1957: Simon's optimism

"It is not my aim to surprise or shock you—but the simplest way I can summarize is to say that there are now in the world machines that think, that learn and that create.

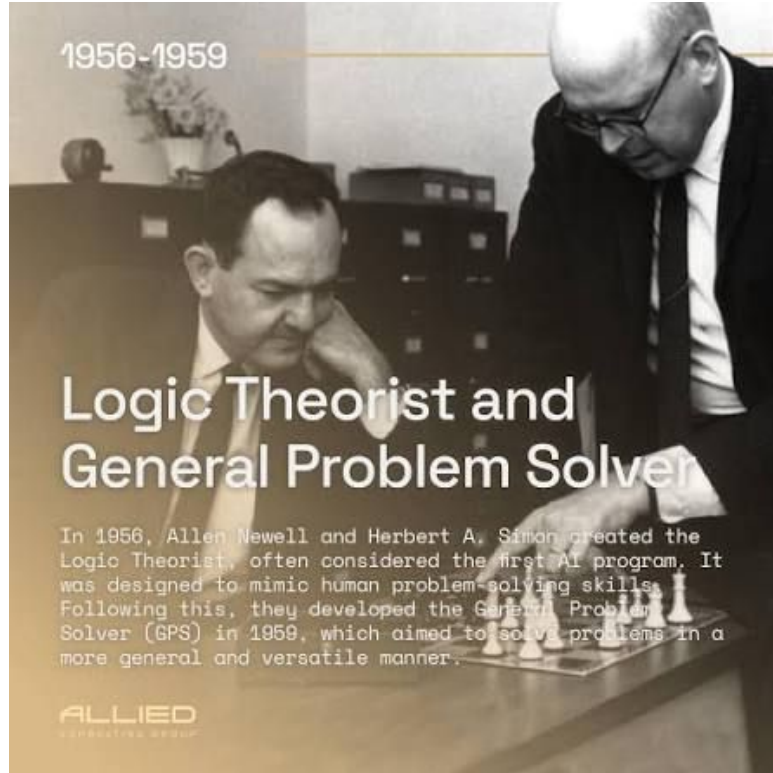
Moreover, their ability to do these things is going to increase rapidly until—in a visible future—the range of problems they can handle will be coextensive with the range to which the human mind has been applied."

1959: Arthur Samuel coins the term “machine learning”



Checker-playing program (Draughts)

1956-9: Alan Newell & Herb Simon (Symbolic AI)

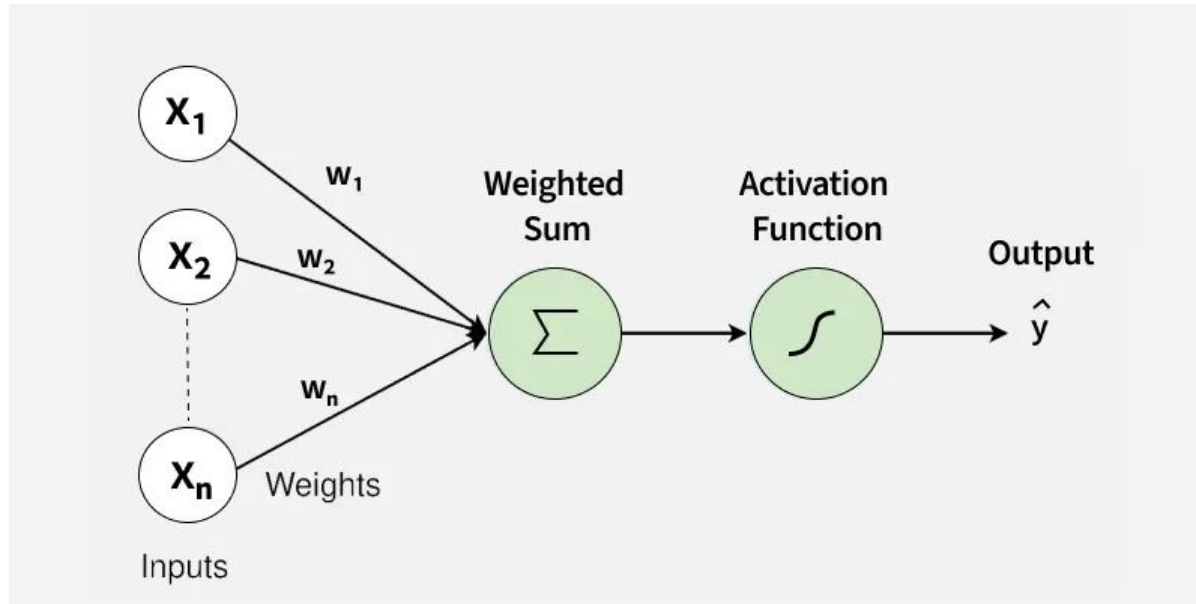


Means-end Analysis

apply operators to minimize the difference between the goal state and the current state

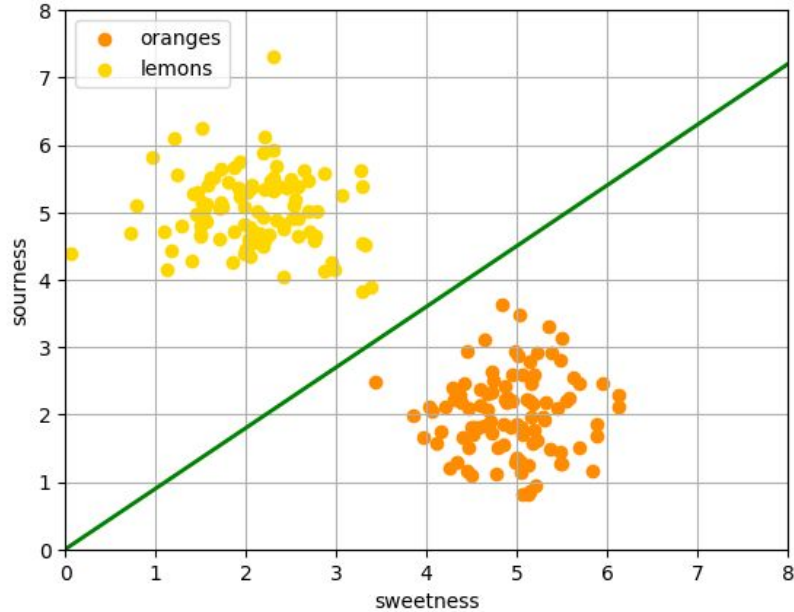
1943: McCullough-Pitts Artificial Neuron

1957: Frank Rosenblatt's Perceptron Algorithm

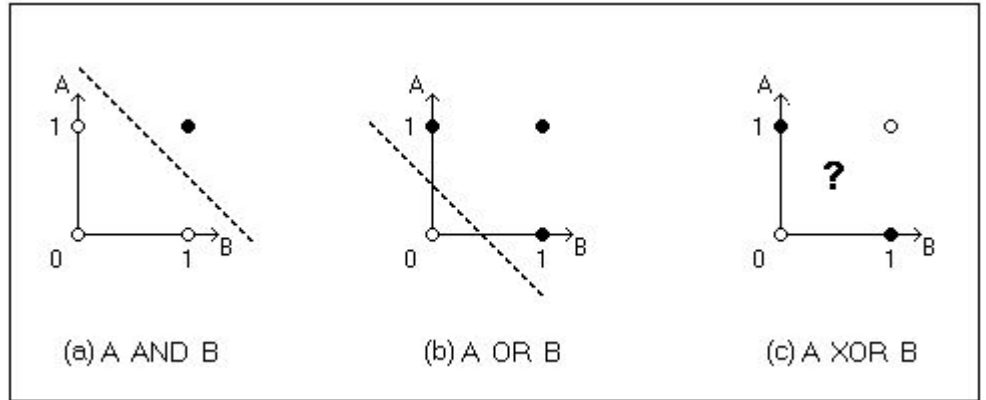


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1969: AI Winter Descends (Minsky & Papert)



[Source](#)



1977: Rule-based systems (Symbolic AI)

- Mycin, an expert system
 - Logic, rules, inference
- Weizenbaum's ELIZA, the therapist, the first chatterbot (chatbot)
 - Pattern matching, substitution rules

User: My wife says I'm often depressed.

ELIZA: I'm sorry to hear you are depressed.

User: The truth is I am quite unhappy.

ELIZA: Do you think coming here will help you not to be unhappy?

User: Well, perhaps by talking to you I will find out why I hate my brother.

ELIZA: Tell me more about your family.

1977: Rule-based systems (Symbolic AI)

- Mycin, an expert system (Ed Feigenbaum)
 - Logic, rules, inference
- Weizenbaum's ELIZA, the therapist, the first chatterbot (chatbot)
 - Pattern matching, substitution rules

User: I was reading about the work of Mother Teresa today.

ELIZA: Tell me more about your family.

1986: Multilayer Perceptron (Neural Network)

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Letter | Published: 09 October 1986

Learning representations by back-propagating errors

[David E. Rumelhart](#), [Geoffrey E. Hinton](#) & [Ronald J. Williams](#)

[Nature](#) **323**, 533–536 (1986) | [Cite this article](#)

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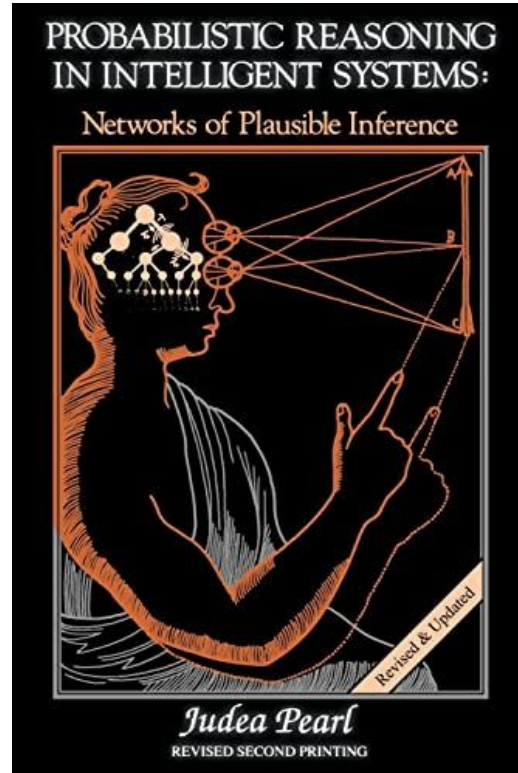
264k Accesses | **27k** Citations | **824** Altmetric | [Metrics](#)

Abstract

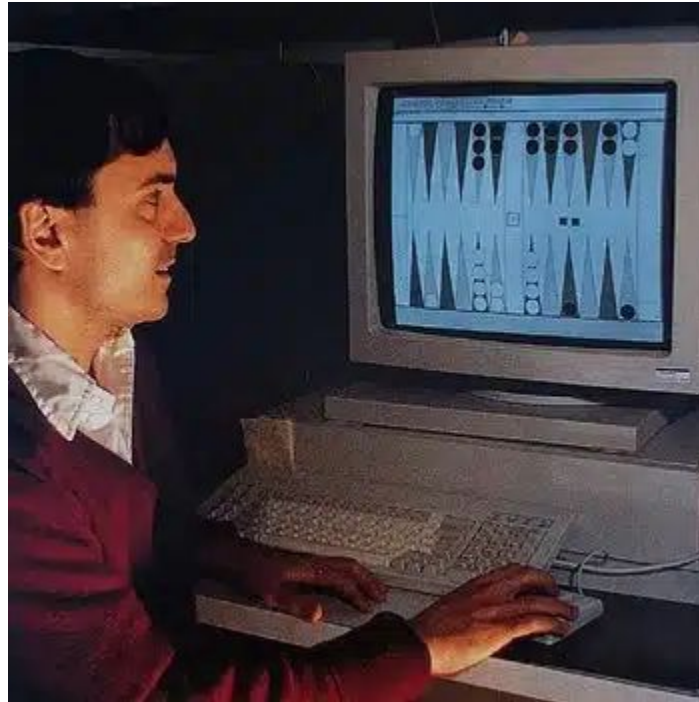
We describe a new learning procedure, back-propagation, for networks of neurone-like units. The procedure repeatedly adjusts the weights of the connections in the network so as to minimize a measure of the difference between the actual output vector of the net and the desired output vector. As a result of the weight adjustments, internal 'hidden' units which are not part of the input or output come to represent important features of the task domain, and the regularities in the task are captured by the interactions of these units. The ability to create useful new features distinguishes back-propagation from earlier, simpler methods such as the perceptron-convergence procedure¹.

Universal
Approximation
Theorem
(1989)

1988: Judea Pearl & the Probabilistic Revolution



1992: IBM's TD-Gammon (MLPs + Reinforcement Learning)



1997: IBM's Deep Blue defeats grandmaster Gary Kasparov



2011: IBM's Watson wins Jeopardy!



2016: DeepMind's AlphaGo defeats grandmaster Lee Sodol



2017: Google's Transformer Architecture

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

She sat on the **bank** of the river.

She went to the **bank** to deposit money.

2022: Large Language Models



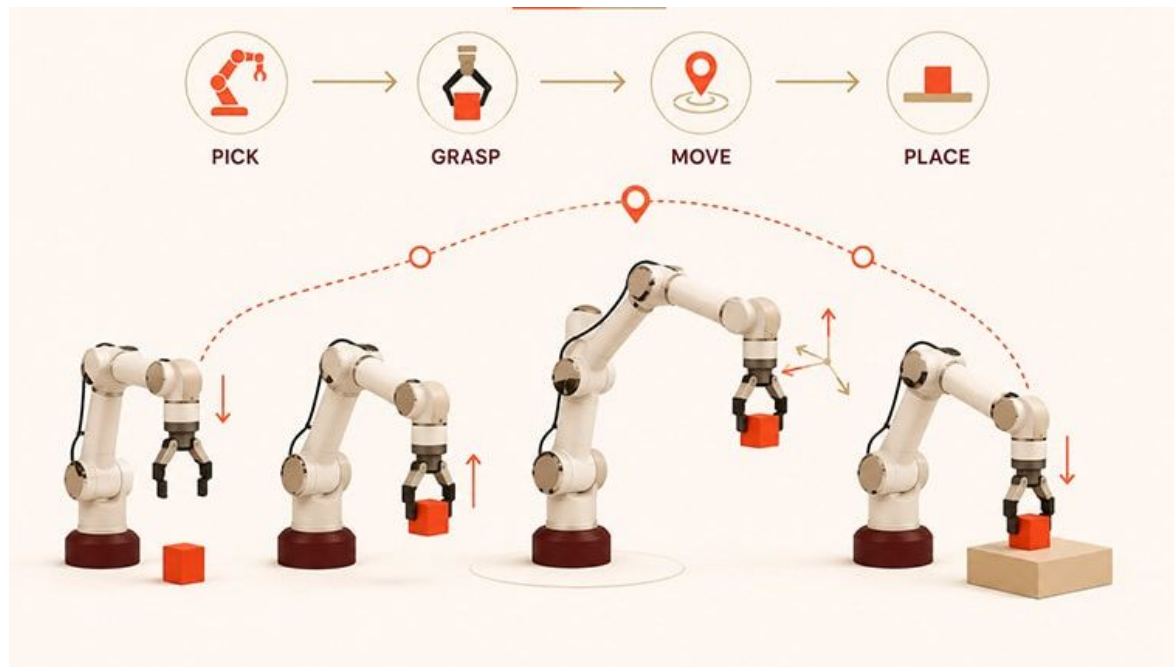
ChatGPT



DeepMind's Alpha Revolution (NNs + RL)

- AlphaGo
- AlphaFold
 - 2: Predict 3D protein structures from amino acid sequences
 - 3: Predict complex structures of proteins, DNA, RNA, etc.
 - Nobel prize (2024)!
- AlphaProof + AlphaGeometry
 - International Math Olympiad Silver Medal (2024)

Multimodal (e.g., Vision Language Action) Models



Multidisciplinary

- Philosophers + Psychologists
 - The mind as an information-processing machine
 - Knowledge is encoded in our minds
 - Thought is used to make decisions
- Mathematicians
 - Logic + Probability
- Economists
 - Utility-maximization
- Neuroscientists + Linguists
- Engineers

Early Warnings

It seems probable that once the machine thinking method had started, it would not take long to outstrip our feeble powers. . . . At some stage therefore we should have to expect the machines to take control, in the way that is mentioned in Samuel Butler's *Erewhon*.

Alan Turing, 1951

If we use, to achieve our purposes, a mechanical agency with whose operation we cannot interfere effectively . . . we had better be quite sure that the purpose put into the machine is the purpose which we really desire.

Norbert Weiner, 1960

It's all fun and games until somebody loses an eye!

- Economic risks
 - Like many technical innovations before it, AI is destined to disrupt the labor market
- Biosecurity risks
 - AI trained on biological or chemical data can be applied to design novel toxins
- Environmental risks
 - Water usage
- Cascading failures
 - Swarms of AI agents executing trades, controlling devices, navigating autonomous vehicles, etc.
- Degradation of trust in media, public institutions, etc.
 - AI can clone voices, create fake videos, etc.
- Autonomous weapons
- Disposability risk
 - AI outsmarts humans, and deems us dispensable!

Course Logistics

CSCI 0410 = CSCI 1411

- They are the exact same course (same lectures, same assignments)!
- 1411 is for graduate students, only!
- Ugrads must register for 410

Prerequisites

- One of CSCI 0190 or 0200
- One of CSCI 0220, MATH 1530, APMA 1650, 1655, CSCI 1450 or 0450
- One of CSCI 0530*, MATH 0520*, 0540* or APMA 0260*

*can be taken concurrently

If you have not yet fulfilled the probability and statistics prerequisite, please do not request an override. It will not be granted. Instead, please fulfill this prerequisite, and come back next semester.

If you have taken linear algebra elsewhere, you must apply to the college for placement notation, which indicates that the prerequisite has been satisfied, and then your override can be granted.

Course Structure

- Lectures M/W/F
 - Lectures are not recorded
 - Lecture notes are posted
- Weekly HW Assignments
 - Assessed via in-person code reviews
- Weekly discussion sections
 - Socially Responsible Computing
 - Conceptual (relate to HWs)
- Two midterms
 - October 9
 - November 9
- Final project (three parts)
 - Final part due December 14, followed by an in-person code review

Getting Started

- Onboarding form
 - Sign up for section
- HW 1 released tomorrow
 - Due Sunday, September 20 at 5pm
- HW 2 released next Tuesday
 - Also due Sunday, September 20 at 5pm

Homeworks are generally posted on Tuesdays and due the following Monday night at 11:59 pm

Shopping Period

- Complete HWs 1 + 2 (due September 20)
- Attend discussion section next week
 - You cannot make up a missed section
 - The lowest section grade is dropped

Grading Breakdown (subject to change)

Assignment	Percentage
Homeworks	20%
Discussion Sections	15%
Midterm 1	25%
Midterm 2	25%
Final Project	15%

To pass the course, you must pass at least one of the midterms.

Collaboration Policy: Working with Peers

It is useful when working on assignments to bounce ideas off one another, and we encourage you to do so.

- You can discuss how to approach a problem with others, using a whiteboard or paper, or even a napkin. But you cannot walk away from these discussions with any artifacts: : no pictures of the whiteboard, no written notes, no napkins.
- Submit only your own work.
- Do not share code or solutions.
- List the names of your collaborators in your README.

Collaboration Policy: Working with Generative AI

We encourage students to use generative AI, but to be careful not to rely on them too heavily, lest they interfere with the learning process.

The programming assignments in CS 410/1411 proceed from TA-authored stencil code to partial scaffolding to almost none at all, so that students can learn to make effective use of generative AI as a coding assistant. The AI-use policy will likewise evolve over the course of the semester.

As a rule of thumb, you are allowed to ask a chatbot to answer questions whose solutions can be found in the textbook.

You are not allowed to ask a chatbot to answer any questions on a homework assignment for you.

Violations of the Collaboration Policy

We take suspected violations of the course collaboration policy seriously!

- Your handin will be flagged if your code passes all our tests, but you fail your code review
- If a second handin is also flagged, you will receive a 0 on both flagged handins, and all homeworks in between
- In addition, these handins may be forwarded to the Academic Code Committee for review

If you have any questions about this policy, please ask the course staff for clarification. *Not understanding our policy is not grounds for not abiding by it.*

Questions?

— (about course logistics) —

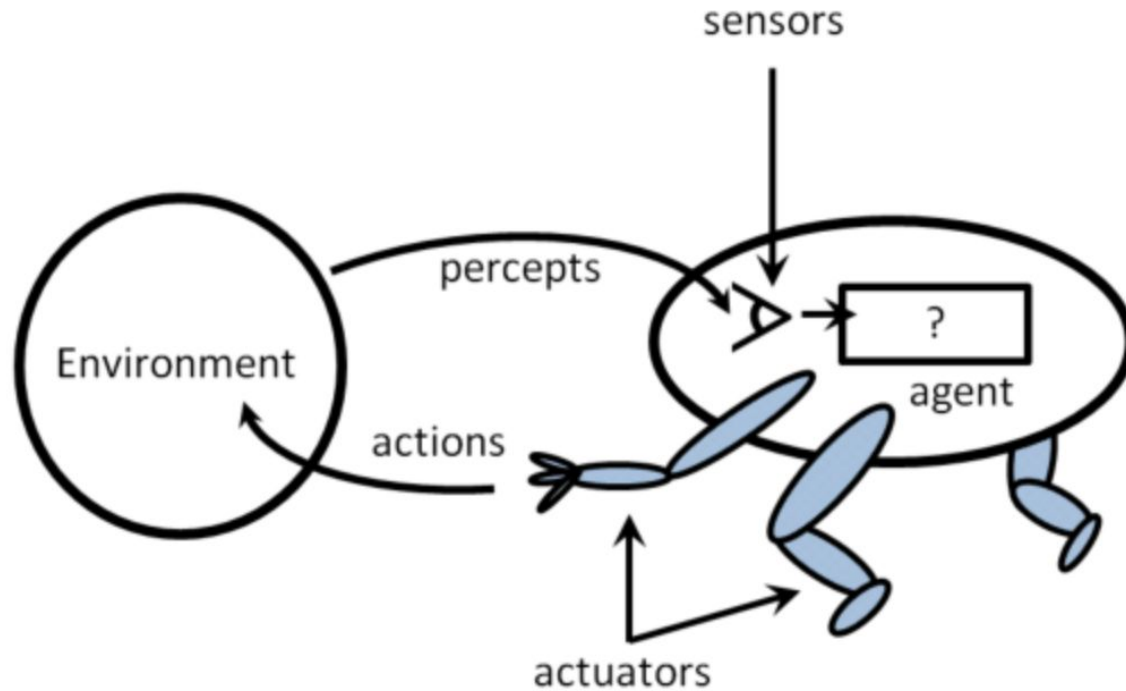
What is AI?

What is an intelligent entity?

	Human	Rational
Think	Thinking humanly: GPS	Thinking rationally: Logicism
Act	Acting humanly: Turing Test	Acting rationally: Modern AI

Table 1: Four approaches to agent design. Following Russell and Norvig, 2000.

What is an autonomous agent?



What characterizes an agent environment?

- Deterministic vs. Nondeterministic
- Discrete vs. Continuous
- Static vs. **Dynamic**
- One-shot vs. **Sequential**
- Single agent vs. **Multiagent**

My research concerns the interactions of multiple autonomous AI agents, typically in dynamic, sequential environments. (The GLAMOUR lab.)

Course Outline

Symbolic AI

- Search & Planning
- Constraint Satisfaction
- Discrete Optimization

Neural Networks

- Continuous Optimization
- Supervised Learning

Modern AI

- Reinforcement Learning
- Unsupervised Learning

